

Лабораторная работа №9

Использование протокола STP. Агрегирование каналов

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Изучить возможности протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.

1. Сформировать резервное соединение между коммутаторами msk-donskayasw-1 и msk-donskaya-sw-3.
2. Настроить балансировку нагрузки между резервными соединениями.
3. Настроить режим Portfast на тех интерфейсах коммутаторов, к которым подключены серверы.
4. Изучить отказоустойчивость резервного соединения.
5. Сформировать и настроить агрегированное соединение интерфейсов Fa0/20 – Fa0/23 между коммутаторами msk-donskaya-sw-1 и msk-donskaya-sw-4.
6. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

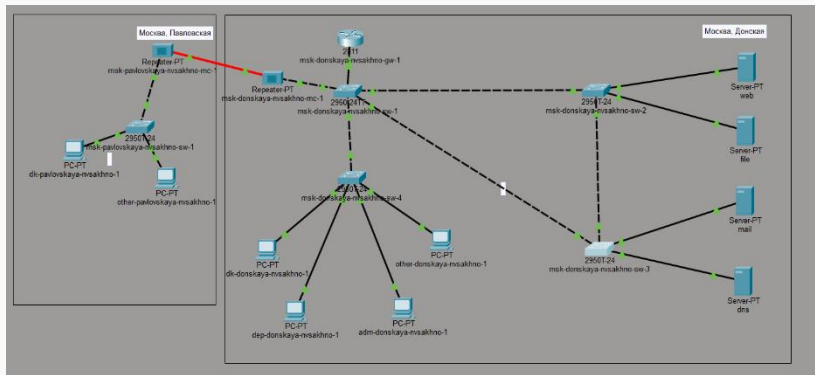


Рис. 1: Логическая схема локальной сети с резервным соединением

```
msk-donskaya-nvsakhno-sw-3(config)#int g0/2
msk-donskaya-nvsakhno-sw-3(config-if)#switchport mode trunk
msk-donskaya-nvsakhno-sw-3(config-if)#exit
msk-donskaya-nvsakhno-sw-3(config)#
```

Рис. 2: Настройка trunk-порта на интерфейсе Gig0/2 коммутатора msk-donskaya-sw-3

```
C:\>ping www.donskaya.rudn.ru

Pinging 10.128.0.10 with 32 bytes of data:

Reply from 10.128.0.10: bytes=32 time=10ms TTL=127
Reply from 10.128.0.10: bytes=32 time=12ms TTL=127
Reply from 10.128.0.10: bytes=32 time=10ms TTL=127
Reply from 10.128.0.10: bytes=32 time=11ms TTL=127

Ping statistics for 10.128.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 12ms, Average = 10ms

C:\>ping mail.donskaya.rudn.ru

Pinging 10.128.0.12 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.12: bytes=32 time=10ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=16ms TTL=127

Ping statistics for 10.128.0.12:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 16ms, Average = 12ms
```

Рис. 3: Пингование сервера mail и web

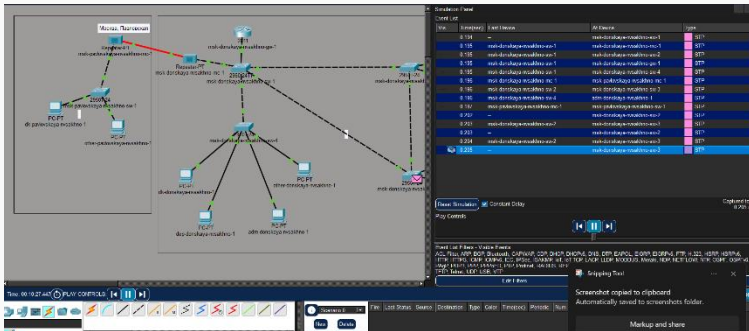


Рис. 4: Режим симуляции движения пакетов ICMP

```
msk-donskaya-nvsakhno-sw-2#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0001.9639.4C79
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32771  (priority 32768 sys-id-ext 3)
             Address     0060.4769.301A
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/2                    Desg FWD 19        128.2    P2p
Fa0/1                    Desg FWD 19        128.1    P2p
Gi0/1                    Root FWD 4         128.25   P2p
Gi0/2                    Desg FWD 4         128.26   P2p

msk-donskaya-nvsakhno-sw-2#
```

Рис. 6: Просмотр состояния протокола STP для vlan 3


```
msk-donskaya-nvsakhno-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-nvsakhno-sw-1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (101), with msk-donskaya-nvsakhno-sw-3 FastEthernet0/3 (1).

msk-donskaya-nvsakhno-sw-1(config)#show spanning-tree vlan 3
^
% Invalid input detected at '^' marker.

msk-donskaya-nvsakhno-sw-1(config)#exit
msk-donskaya-nvsakhno-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-nvsakhno-sw-1#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    24579
             Address     0001.9639.4C79
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    24579 (priority 24576 sys-id-ext 3)
             Address     0001.9639.4C79
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1 Desg FWD 19 128.1 P2p
Gi0/1 Desg FWD 4 128.25 P2p
Fa0/24 Desg FWD 19 128.24 P2p
Gi0/2 Desg FWD 4 128.26 P2p

msk-donskaya-nvsakhno-sw-1#
```

Рис. 7: Настройка коммутатора msk-donskaya-sw-1 корневым

Выполнение лабораторной работы

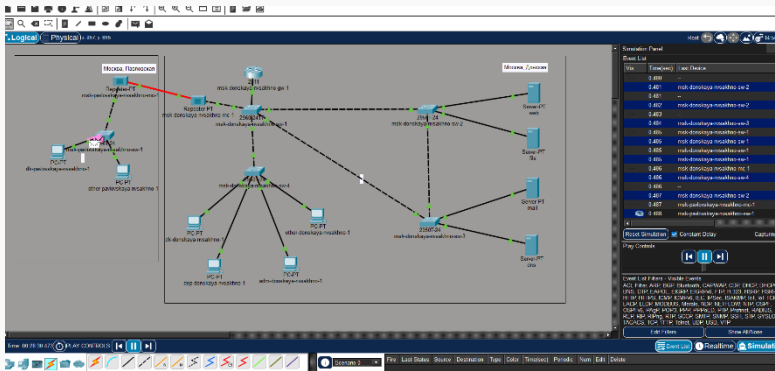


Рис. 8: Режим симуляции движения пакетов ICMP к серверу web

```
msk-donskaya-nvsakhno-sw-1(config)#interface f0/1
msk-donskaya-nvsakhno-sw-1(config-if)#spanning tree portfast
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-nvsakhno-sw-1(config-if)#spanning tree portfast
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-nvsakhno-sw-1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-nvsakhno-sw-1(config-if)#|
```

Рис. 10: Настройка режима Portfast

Выполнение лабораторной работы

```
C:\>ping -n 1000 mail.donskaya.rudn.ru
Pinging 10.128.0.12 with 32 bytes of data:

Reply from 10.128.0.12: bytes=32 time=52ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=12ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=10ms TTL=127
Reply from 10.128.0.12: bytes=32 time=22ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=13ms TTL=127
Reply from 10.128.0.12: bytes=32 time=35ms TTL=127
Reply from 10.128.0.12: bytes=32 time=16ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=10ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=12ms TTL=127
Reply from 10.128.0.12: bytes=32 time=10ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=14ms TTL=127
Reply from 10.128.0.12: bytes=32 time=10ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=12ms TTL=127
Reply from 10.128.0.12: bytes=32 time=10ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=12ms TTL=127
Reply from 10.128.0.12: bytes=32 time=12ms TTL=127
Reply from 10.128.0.12: bytes=32 time=23ms TTL=127
Reply from 10.128.0.12: bytes=32 time=11ms TTL=127
Reply from 10.128.0.12: bytes=32 time=13ms TTL=127
```

Рис. 11: Пингование mail.donskaya.rudn.ru

```
msk-donskaya-nvsakhno-sw-3(config-if)#exit
msk-donskaya-nvsakhno-sw-3(config)#int g0/2
msk-donskaya-nvsakhno-sw-3(config-if)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
msk-donskaya-nvsakhno-sw-3(config-if)#
```

Рис. 12: Разрыв соединения

```
C:\>ping mail.donskaya.rudn.ru

Pinging 10.128.0.12 with 32 bytes of data:

Reply from 10.128.0.12: bytes=32 time=47ms TTL=127
Reply from 10.128.0.12: bytes=32 time=22ms TTL=127
Reply from 10.128.0.12: bytes=32 time=14ms TTL=127
Reply from 10.128.0.12: bytes=32 time=28ms TTL=127

Ping statistics for 10.128.0.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 47ms, Average = 27ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Рис. 13: Время восстановления соединения

Выполнение лабораторной работы

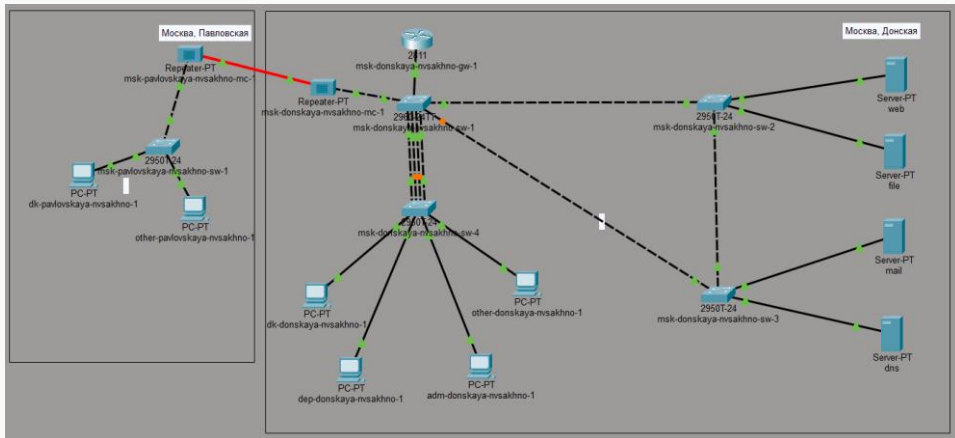


Рис. 18: Логическая схема локальной сети с агрегированным соединением

```
msk-donskaya-nvsakhno-sw-1(config)#int f0/23
msk-donskaya-nvsakhno-sw-1(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (101), with msk-donskaya-
nvsakhno-sw-3 FastEthernet0/3 (1).

msk-donskaya-nvsakhno-sw-1(config-if)#no switchport mode trunk
msk-donskaya-nvsakhno-sw-1(config-if)#
```

Рис. 19: Настройка агрегирования каналов на msk-donskaya-nvsakhno-sw-1


```
msk-donskaya-nvsakhno-sw-4(config)#int range f0/20 - 23
msk-donskaya-nvsakhno-sw-4(config-if-range)#no switchport access vlan 104
msk-donskaya-nvsakhno-sw-4(config-if-range)#exit
msk-donskaya-nvsakhno-sw-4(config)#interface range f0/20 - 23
msk-donskaya-nvsakhno-sw-4(config-if-range)#channel group 1 mode on
% Ambiguous command: "channel group 1 mode on"
msk-donskaya-nvsakhno-sw-4(config-if-range)#channel group 1 mode on
% Ambiguous command: "channel group 1 mode on"
msk-donskaya-nvsakhno-sw-4(config-if-range)#channel-group 1 mode on
msk-donskaya-nvsakhno-sw-4(config-if-range)#
Creating a port-channel interface Port-channel 1

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (dtp mode of Fa0/23 is on,
Fa0/20 is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (dtp mode of Fa0/23 is on,
Fa0/21 is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (dtp mode of Fa0/23 is on,
Fa0/22 is off )

msk-donskaya-nvsakhno-sw-4(config-if-range)#e
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (101), with msk-donskaya-
nvsakhno-sw-1 FastEthernet0/3 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (101), with msk-donskaya-
nvsakhno-sw-1 FastEthernet0/5 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (101), with msk-donskaya-
nvsakhno-sw-1 FastEthernet0/4 (1).

msk-donskaya-nvsakhno-sw-4(config)#exit
msk-donskaya-nvsakhno-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-nvsakhno-sw-4#interface port-channel 1

% Invalid input detected at '^' marker.

msk-donskaya-nvsakhno-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-nvsakhno-sw-4(config)#interface port-channel 1
msk-donskaya-nvsakhno-sw-4(config-if)#switchport mode trunk
msk-donskaya-nvsakhno-sw-4(config-if)#exit
msk-donskaya-nvsakhno-sw-4(config)#
```

Рис. 20: Настройка агрегирования каналов на msk-donskaya-nvsakhno-sw-1

В результате выполнения лабораторной работы я изучил возможности протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.